Mustafa Çağatay (Chaatai) Karakan
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A mechanical/biomedical engineer with over a decade of interdisciplinary research experience spanning microfabrication, microphysiological systems for tissue engineering, and bioinspired device design. Proven expertise in developing microfluidic and organ-on-chip platforms, high-resolution 3D-printed scaffolds, and micro/nanoelectromechanical systems for biosensing applications. Adept at leading and collaborating in cross-disciplinary projects from concept to fabrication, integrating mechanical engineering principles with cell biology, microfluidics, and precision metrology. Published extensively in high-impact journals (Google Scholar link), and experienced in mentoring and collaborating with diverse research teams, while managing cutting-edge laboratory facilities.

Education

Boston University **Boston, MA**
Ph.D. in Mechanical Engineering, GPA: 4.0/4.0 *09/2017 - 01/2023*

- Thesis: A Direct-Laser-Written Heart-on-a-Chip Platform for Generation and Stimulation of Engineered Heart Tissues
- Fellow of NSF CELL-MET Engineering Research Center

Bilkent University **Ankara, Turkey**
B.S. and M.S. in Mechanical Engineering, GPA: 3.64/4.0 *2009–2013, 2014–2016*

- Thesis: Development of Functionalized Nanoelectromechanical Systems Functionalized by Vacuum Organic Thermal Evaporation for Gas Sensing Applications

Work Experience

Boston University and Wyss Institute at Harvard University **Boston, MA**
Postdoctoral Researcher, Advisor: Prof. Christopher Chen *02/2023 - Present*

- Supervising the Multiscale Laser Lithography Laboratory, training and mentoring new users, and using the lab for various cell patterning (topographical cues and microcontact printing), tissue engineering (high-precision fabrication of molds and scaffolds), MEMS (building electrostatic lenses and polymeric resonators on a chip), micro biorobotics (magnetically and acoustically controllable microrobots), and optics (microlenses) applications.
- Developing microphysiological systems with electromechanical stimulation capabilities to reliably grow and mature human-engineered 3D heart tissues on chips, with tunable sizes and geometries. Increased the throughput of the platform by 4×, and enabled long-term (>1 month) culturing.
- Tested the potential of direct-laser-written metamaterial scaffolds to grow human-engineered heart tissues for in vitro and translational applications, successfully generated and evaluated beating heart patches on centimeter-scale scaffolds.

Boston University, Department of Mechanical Engineering **Boston, MA**
Graduate Researcher, Advisors: Prof. Kamil Ekinci and Prof. Alice White *09/2017 - 01/2023*

- Developed a microfluidic tissue stretcher to apply forces on engineered tissues. Further engineered the platform to grow and mature functional 3D cardiac microbundles in various lengths and geometries.
- Built a finite element model (FEM) of the heart-on-a-chip platform to simulate the elastic response of the scaffold upon applied pressure or tissue contractions.
- Developed a microfluidic setup to test and characterize 3D-printed heart valves with unprecedented (<1 Pa) pressure sensitivity.
- Measured and correlated the pore sizes of the extracellular matrices with varying degrees of density.
- 3D printed micron scale, biocompatible structures for cell patterning and targeted drug delivery applications.
- Developed a method to detect living bacteria inside a hanging droplet using the displacement fluctuations of an Atomic Force Microscope (AFM) cantilever.

- Designed, fabricated, and tested the following microfluidic devices with integrated electrodes and impedance measurement capabilities, using the four-probe technique through a lock-in amplifier:
  - An organ-on-a-chip platform that allows excitation and high-throughput detection of contractile stresses exerted by the engineered heart tissues.
  - A microfluidic device for trapping the red blood cells and characterizing their viability based on their membrane fluctuations.
  - A Coulter Principle-based flow monitoring sensor to track and characterize the biological and particulate microflows.
  - An antibiotic susceptibility testing (AST) device that identifies the right antibiotic treatment in an hour by measuring bacterial motion and growth.

Bilkent University **Ankara, Turkey**
Graduate Researcher, Advisor: Prof. M. Selim Hanay *2014 - 2016*

- Designed, fabricated, and characterized multi-mode nanoelectromechanical systems (NEMS) with electrothermal actuation and piezoresistive read-out.
- Optimized procedures to chemically functionalize NEMS for sensing applications by organic thin film evaporation and micropipette-assisted drop-casting of polymers.
- Designed and built a gas bubbler-based delivery system that generates a tunable concentration of volatile organic compounds. Detected potential cancer biomarkers with a functionalized NEMS sensor.
- Built an electrospray ionization system to deliver nanoparticles to NEMS sensors.
- Developed a mode-coupling-based technique for the detection of nanomechanical vibration modes.

Technical Skills

Prototyping, Microfabrication & Surface Functionalization: Multiscale 3D Printing (two-photon direct laser writing (Nanoscribe and UpNano), Stereolithography (Formlabs), Photolithography, E-Beam Lithography, Soft Lithography, PDMS microfabrication and molding, surface functionalization and thin-film deposition (thermal and e-beam evaporation, sputter coating, silane deposition).

Cell biology: Bacteria and mammalian cell culture, stem cell culture and hiPSC differentiation, hydrogels (collagen, fibrin, matrigel), organoid and 3D tissue generation, immunostaining.

Scientific Characterization & Precision Metrology: Scanning Electron Microscopy, Confocal Microscopy, Atomic Force Microscopy, Profilometry.

Electronics: Circuits, Oscilloscopes and Function Generators, Lock-in Amplifiers and Phase Sensitive Detection, Data Acquisition, Signal Processing and Spectral Analysis, PIC, Arduino.

Software & Computer Languages: SolidWorks, COMSOL Multiphysics, MATLAB, Python, Excel, C++, LabVIEW, OriginPro, ImageJ, Adobe Illustrator & Photoshop, LaTeX.

Leadership and Teaching

Boston University **Boston, MA**
Teaching Fellow *2017 - present*

- Acting as the superuser and laboratory safety coordinator of the Multiscale Laser Lithography Lab.
- Leading the preparation of the research section of the annual reports of the NSF-funded Engineering Research Center (ERC) on Cellular Metamaterials (CELL-MET) for 4 years.
- Peer-reviewing articles and grant applications in advanced microfabrication, biological tissue engineering, and MEMS fields for high-impact journals (Nature Communications, Advanced Functional Materials, Small, Lab on a Chip, Advanced Healthcare Materials) and universities (ETH Zurich).
- Defined and supervised multiple projects, and mentored trainees for NSF-funded Research Experiences for Undergraduates (REU), Teachers (RET), STEM Pathways, and Boston University Research in Science and Engineering (RISE) programs.

- Mentoring many undergraduate and graduate students from different cultures on microfabrication, finite element simulations, electrical circuits, microscopy, microfluidics, cell culture, and hiPSC differentiation.
- Performed as a senior teaching and lab assistant for ME310-Measurement and Instrumentation for two semesters.
- Volunteered in Cambridge Science Festival: Engineering Biology to Solve Important Problems in Medicine, Energy, and the Environment (2022).

Bilkent University **Ankara, Turkey**
Teaching Fellow *2014 - 2016*

- Acted as a liaison between students and lead engineers for industry-funded senior projects.
- Performed as a teaching and lab assistant for every semester:
  - ME384 - Mechatronic Systems *Spring 2014, Spring 2015, Spring 2016*
  - ME341 - Dynamics and Control I *Fall 2016*
  - ME481 - Mechanical Engineering Design I *Fall 2015*

Scholarships and Awards

- 2nd place award in NSF CELL-MET Perfect Pitch Competition (2026).
- National Interest Waiver (NIW) approval from the U.S. government for exceptional contributions and national importance of my work (2024).
- 1st place award in NSF CELL-MET Perfect Pitch Competition (2023).
- Fellow of NSF CELL-MET Engineering Research Center in Cellular Metamaterials (2018-present).
- 1st place award in Boston University Mechanical Engineering Scientific Image Contest (2022).
- 2nd place award in NSF CELL-MET Perfect Pitch Competition (2020).
- Tuition waiver and Distinguished Departmental Fellowship from Boston University (2017-2022).
- Tuition waiver and Financial Aid – Bilkent University Mechanical Engineering Graduate Program (2014-2017).
- Best Poster Award – Bilkent Mechanical Engineering Department Graduate Student Conference (2015).
- 50% Merit Scholarship – Bilkent University (2009 – 2013).
- Honor Certificate – Bilkent University (2011 – 2012).
- History of Turkey, Honor Certificate – Bilkent University (2012).
- Best Project Award – ME102 Introduction to Systems Engineering, Bilkent University (2010).

Publications and Conference Presentations

M. Ç. Karakan (2026) "3D-Printed Porous Anchors for Suspended Tissue Engineering" Engineering in Cardiovascular Medicine. Ann Arbor, MI (poster).

M. Ç. Karakan (2026) "Harnessing Direct Laser Writing for Cardiac Tissue Engineering, and a Tutorial on Aligned 3D Printing" *Nanoscribe User Meeting*. Los Angeles, CA (invited talk).

A. Vasan, Q. Nguyen, E. Davis, **M. Ç. Karakan**, E., Westphal, V. Shah, W. Wong, E. Lejeune, J. Eyckmans (2026). "A high-throughput, 3D microtissue platform for multiparametric analysis of tissue remodeling". *bioRxiv*.

N. Farid, S. Kianersi, A. Sharif, A. Daly, **M. Ç. Karakan**, C. S. Chen, G. M. O'Connor (2026). "Large-Area Flexible Photopolymerized Scaffolds: Fabrication and Application to Cardiomyocytes". *ACS Applied Materials & Interfaces*.

J. Ewoldt, S. DePalma, M. Jewett, **M. Ç. Karakan**, Y. Lin, P. Hashemian, X. Gao, L. Lou, M. McLellan, J. Tabares, M. Ma, A. S. Coariti, J. He, K. Toussaint, T. Bifano, S. Ramaswamy, A. White, A. Agarwal, E. Lejeune, B. Baker, C.S. Chen (2025). "Stem cell-derived cardiomyocyte in vitro models: benchmarking current progress to inform improved tissue maturation". *Nature Methods*.

S.Cherukumilli, F. C. Kirmizitas, D. P. Rivas, M. Sokolich, **M. Ç. Karakan**, A. E. White, S. Das (2024). "Programmable Acoustic Modular Microrobots". *Journal of Micro and Bio Robotics*.

M. Ç. Karakan, J. Ewoldt, A. Segarra, S. Sundaram, A. E. White, C. S. Chen, K. L. Ekinci (2024). "Geometry and Length Control of 3D Engineered Heart Tissues Using Direct Laser Writing". *Lab on a Chip*.

H. Kobeissi, J. Jilberto, **M. Ç. Karakan**, X. Gao, S. J. DePalma, S. L. Das, L. Quach, J. Urquia, B. M. Baker, C. S. Chen, D. Nordsletten, E. Lejeune (2024). "Microbundlecompute: Automated segmentation, tracking, and analysis of sub-domain deformation in cardiac microbundles". *PLOS One*.

S.Cherukumilli, F. C. Kirmizitas, M. Sokolich, A. Valencia, **M. Ç. Karakan**, A. E. White, A.A. Malikopoulos, S. Das (2023). "Programmable Modular Acoustic Microrobots". *IEEE International Conference on Manipulation, Automation and Robotics at Small Scales (MARSS)*.

Y. Yang, F. C. Kirmizitas, M. Sokolich, A. Valencia, D. Rivas, **M. Ç. Karakan**, A. E. White, A.A. Malikopoulos, S. Das (2023). "Rolling Helical Microrobots for Cell Patterning" *IEEE International Conference on Manipulation, Automation and Robotics at Small Scales (MARSS)*.

M. Ç. Karakan, A. B. Ari, M. Kelleci, C. Yanik, I. I. Kaya, O. Tastan, M. S. Hanay (2023). "Vapor Sensing of Colorectal Cancer Biomarkers in Isolation by Bare and Functionalized Nanoelectromechanical Sensors". *IEEE Sensors Journal*.

M. Ç. Karakan (2023). "A Direct-Laser-Written Heart-on-a-Chip Platform for Generation and Stimulation of Engineered Heart Tissues". *Doctoral dissertation, Boston University*.

C. Michas, **M. Ç. Karakan**, P. Nautiyal, J. G. Seidman, C. E. Seidman, A. Agarwal, K. L. Ekinci, J. Eyckmans, A. E. White, C. S. Chen (2022). "Engineering a living cardiac pump on a chip using high precision fabrication". *Science Advances*.

R. K. Jayne*, **M. Ç. Karakan***, K. Zhang*, N. Pierce, C. Michas, D.J. Bishop, C. S. Chen, K. L. Ekinci, A. E. White (2021)."Direct laser writing for cardiac tissue engineering: a microfluidic heart on a chip with integrated transducers". *Lab on a Chip*. *Co-first authors

C. Ti, A. B. Ari, **M. Ç. Karakan**, C. Yanik, I. I. Kaya, M. S. Hanay, O. Svitelskiy, M. González, H. Seren, K. L. Ekinci (2021). "Frequency-Dependent Piezoresistive Effect in Top-down Fabricated Gold Nanoresistors". *Nano Letters*, 21(15), pp.6533-6539.

Y. Kilinc, **M. Ç. Karakan**, Y. Leblebici, M. S. Hanay, B. E. Alaca (2020). "Observation of coupled mechanical resonance modes within suspended 3D nanowire arrays". *Nanoscale*, 12(43), 22042-22048.

J. Tien, U. Ghani, Y. W. Dance, A. J. Seibel, **M. Ç. Karakan**, K. L. Ekinci, and C. M. Nelson (2020). "Matrix pore size governs escape of human breast cancer cells from a microtumor to an empty cavity".*Iscience*, 23(11), 101673.

M. Ç. Karakan and K. L. Ekinci (2019). "Electrical Resistance Fluctuations as a Probe to Monitor Colloidal Flows". *APS March Meeting 2019*, Boston, MA (oral).

A. B. Ari, **M. Ç. Karakan**, C. Yanik, I. I. Kaya, M. S. Hanay (2018). "Intermodal coupling as a probe for detecting nanomechanical modes". *Physical Review Applied*, 9(3), 034024.

M. Ç. Karakan, A. Kirlioglu, K. L. Ekinci (2018). "Detection of Bacterial Movements in a Picoliter-sized Droplet Using a Microcantilever". *APS March Meeting 2018*, Los Angeles, CA (oral).

M. N. Esfahani, Y. Kilinc, **M. Ç. Karakan**, E. Orhan, M. S. Hanay, Y. Leblebici, B. E. Alaca (2018). "Piezoresistive silicon nanowire resonators as embedded building blocks in thick SOI". *Journal of Micromechanics and Microengineering*, 28(4), 045006.

M. Ç. Karakan, A. B. Ari, C. Yanik, I. I. Kaya, M. S. Hanay (2016). "Chemical Functionalization of Nanomechanical

Sensors with Vacuum Thermal Evaporation". 13th International Workshop on Nanomechanical Sensing. Delft, Netherlands (poster).

M. Ç. Karakan, D. Majidi, A. B. Ari, C. Yanik, I. I. Kaya, M. S. Hanay (2015). "Towards Detection of Cancer Biomarkers by NEMS based Electronic Nose". 12th International Workshop on Nanomechanical Sensing. Auckland, New Zealand (poster).